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Experiment #6 Pulse Amplitude Modulation (PAM)

Report #2

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Abstract

This experiment provides a practical exploration of the fundamental principles of Pulse Amplitude Modulation (PAM) using laboratory kits and simulation tools. Key concepts such as pulse train generation, sampling, the Nyquist rate, and aliasing are investigated to show their roles in the conversion of analog signals into digital form. The experiment also examines the influence of sampling pulses and duty cycle variations on signal behaviour in both the time and frequency domains, highlighting their impact on communication system performance. Additionally, Time Division Multiplexing (TDM) is demonstrated, with particular attention to channel crosstalk and system limitations. The results are further supported through simulations using the Cassy Lab software.

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Theory

Pulse Train

A pulse train is a sequence of evenly spaced rectangular pulses that repeat over time. Each pulse has a specific on-time and off-time, forming a periodic pattern used widely in communication and signal-processing systems. An important characteristic of a pulse train is its duty cycle, which affects the shape and behaviour of the signal.[1]

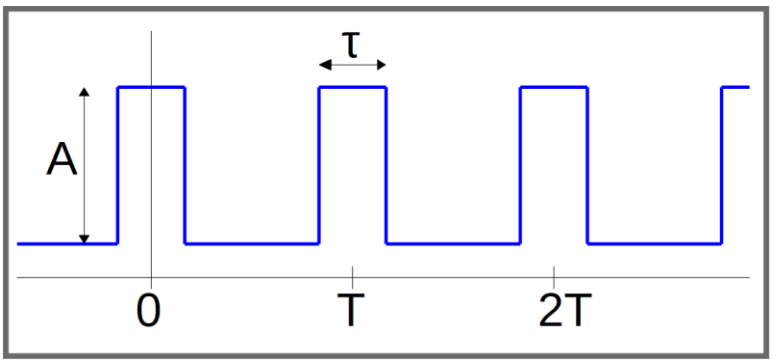


Figure 1: Pulse Train [1]

Duty Cycle

The term "Duty Cycle" refers to the proportion of time that a periodic signal or waveform spends in the active or on state compared to the total time of one complete cycle. It is often expressed as a percentage and is a critical parameter in various applications. [2] It is calculated as:

$$\text{Duty Cycle} = \frac{\text{Pulse On−Time}}{\textit{Total Period}} * 100\%$$

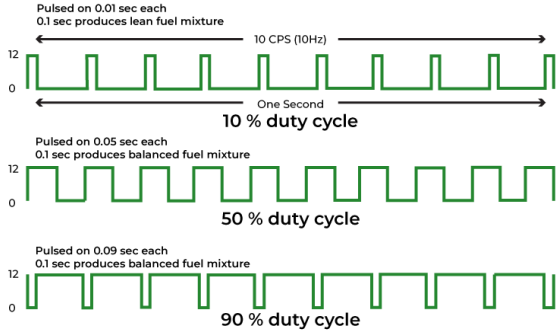


Figure 2: Duty Cycle effect on the pulse train. [2]

Pulse Amplitude Modulation (PAM)

Pulse amplitude modulation is a technique in which the amplitude of each pulse is controlled by the instantaneous amplitude of the modulation signal. It is a modulation system in which the signal is sampled at regular intervals and each sample is made proportional to the amplitude of the signal at the instant of sampling. This technique transmits the data by encoding in the amplitude of a series of signal pulses. [3]

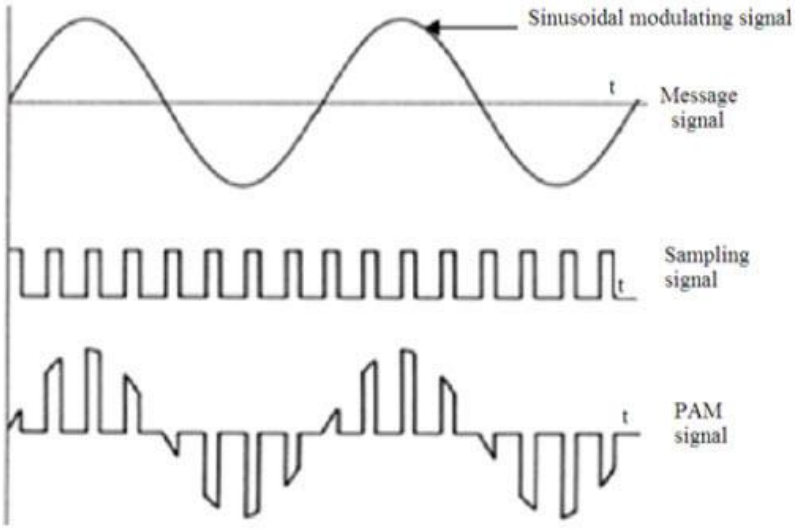


Figure 3: Pulse Amplitude Modulation Signal [3]

Sampling

Sampling is the process by which a continuous time continuous amplitude signal is converted into a discrete time continuous amplitude signal. [4]

Sampling Techniques:

1. **Ideal Sampling** (Note: not used in our experiment because it cannot be generated)

Ideal sampling (also called impulse sampling or instantaneous sampling) is a theoretical sampling technique in which a continuous-time signal is multiplied by a periodic train of Dirac delta impulses, producing a sequence of impulses whose amplitudes equal the instantaneous values of the original signal at the sampling instants. [5]

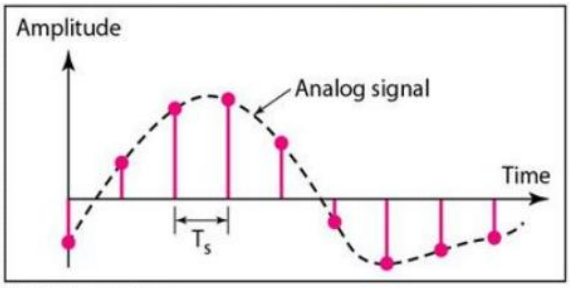


Figure 4: Ideal Sampling [4]

2. Natural Sampling

Natural Sampling is a practical method of sampling in which pulse have finite width equal to τ. Sampling is done in accordance with the carrier signal which is digital in nature.[6]

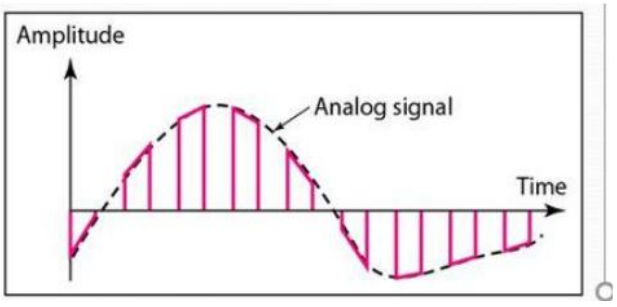


Figure 5: Natural Sampling [4]

3. Flat-Top Sampling

Flat top sampling is like natural sampling i.e; practical in nature. In comparison to natural sampling flat top sampling can be easily obtained. In this sampling techniques, the top of the samples remains constant and is equal to the instantaneous value of the message signal x(t) at the start of sampling process. Sample and hold circuit are used in this type of sampling. .[6]

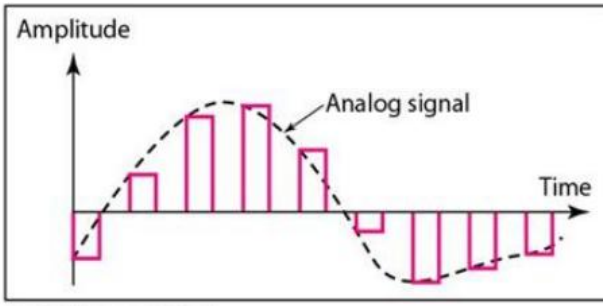


Figure 6: Flat-Top Sampling.[6]

Pulse Amplitude Demodulation

For the demodulation of the PAM signal, the PAM signal is fed to the low pass filter. The low pass filter eliminates the high-frequency ripples and generates the demodulated signal. This signal is then applied to the inverting amplifier to amplify its signal level to have the demodulated output with almost equal amplitude with the modulating signal.[7]

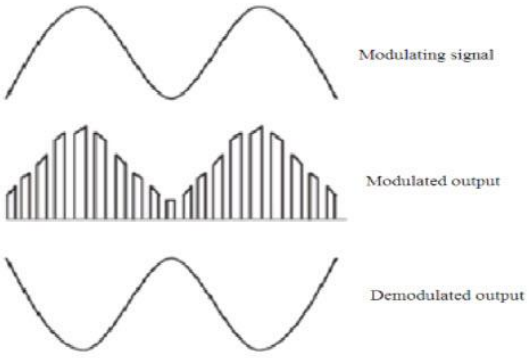


Figure 7: Pulse Amplitude Demodulation [7]

Aliasing

Aliasing is a distortion that occurs in signal processing when a continuous signal is sampled at a rate that is too low to accurately represent it. When the sampling frequency is below the Nyquist rate—defined as twice the highest frequency contained in the signal—the high-frequency components fold back and appear as lower frequencies in the sampled signal, creating false or misleading information. This results in a reconstructed signal that does not match the original.[8]

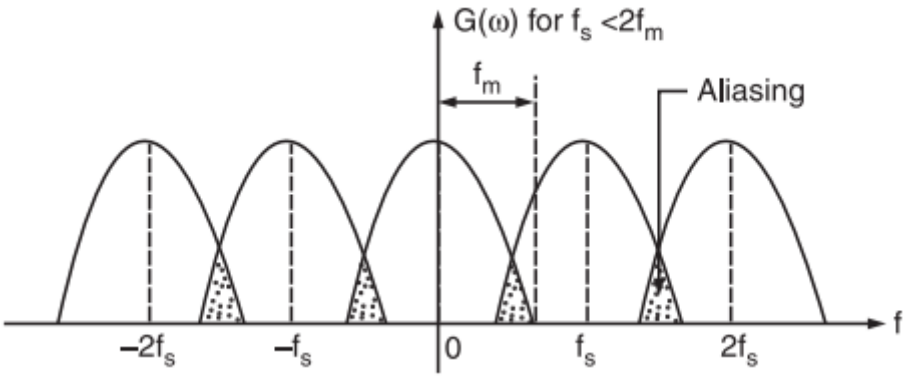


Figure 8: Aliasing [9]

Time Division Multiplexing (TDM)

A technique which allows multiple users to use the same channel by assigning each user a portion of the transmission time without interfering with other users.[4]

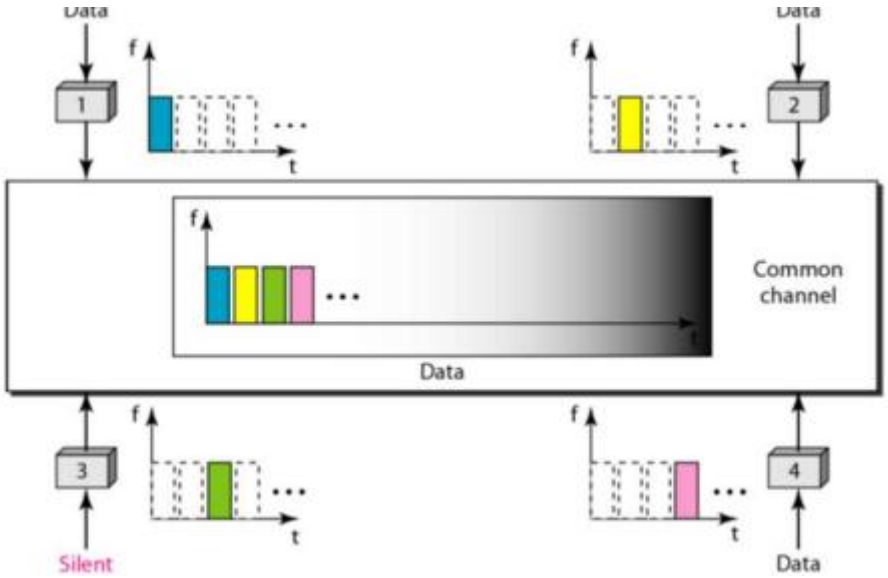


Figure 9: Time Division Multiplexing.[4]

Cross Talk

Crosstalk is defined as interference from other communication channels, which can occur through capacitive or inductive coupling, imprecise timing, or insufficient discrimination between signals. It is classified into near-end and far-end types, and can be intelligible or unintelligible, affecting the error margin in transmission lines.[10]

Procedure

Part 1: Time and Frequency Characteristics of pulse train

The components have been connected as shown in the figure below, A **pulse train** at the function generator has been selected with

Freq = 1 kHz, VSS = 10 V, and duty cycle = 10%



Figure 10: Characteristics of pulse train setup.

Pulse Train in time and frequency domain (D = 10%)

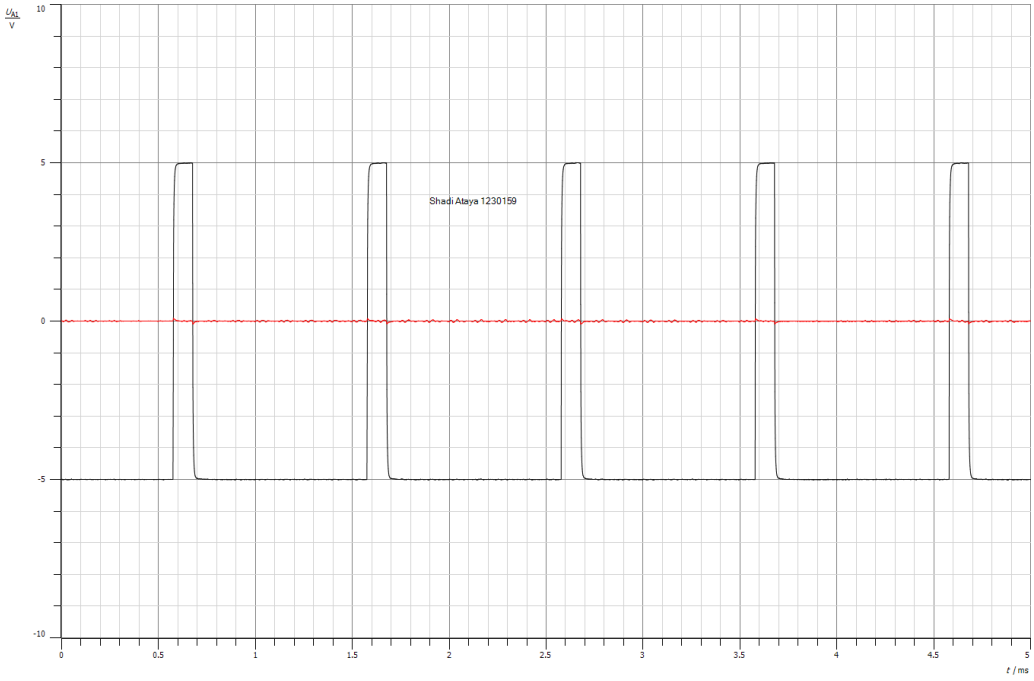


Figure 11: Pulse Train in time domain (D = 10%)

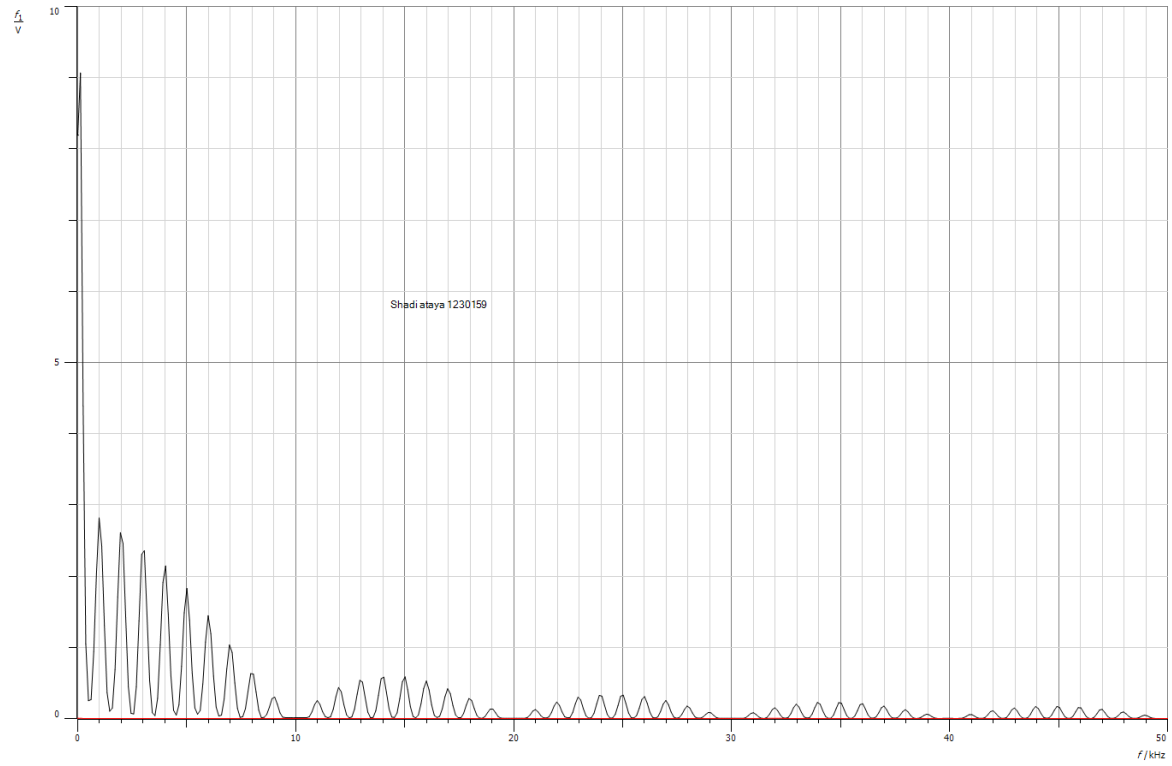


Figure 12: Pulse Train in frequency domain (D = 10%)

Zero crossing when D = 10%

- The zero crossings of the pulse spectrum envelope indicate the frequencies where the spectral amplitude becomes zero. For the previous spectrum, where the duty cycle is 10%, the first zero crossing occurs at 10 kHz. This result can be compared with the theoretical value of the first zero crossing, which can be calculated by

$$F(zerocrossing) = \frac{F(pulse\ train)}{D}$$

by plug the values, the first zero crossing = 1000/0.1 = 10khz this also means that every 10khz there will be a zero crossing in the frequency domain which confirms that the results we got are correct

Zero crossing when D = 20%

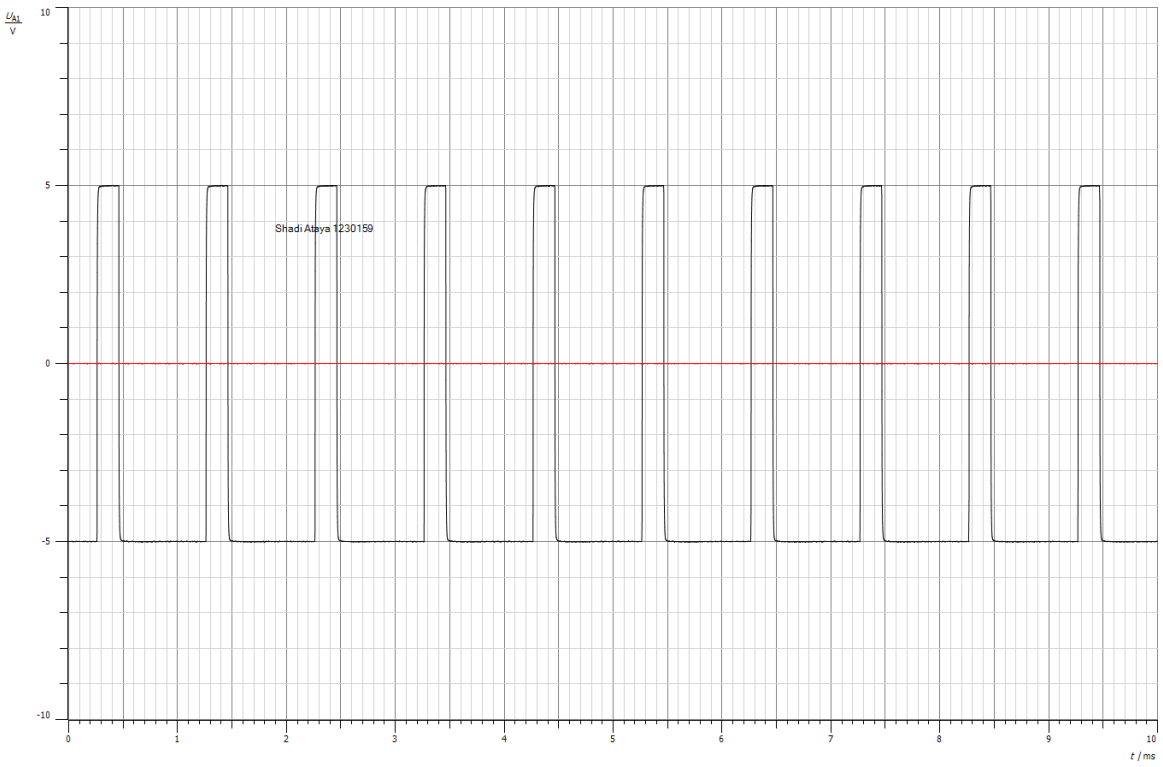


Figure 13 : Pulse Train in time domain (D = 20%)

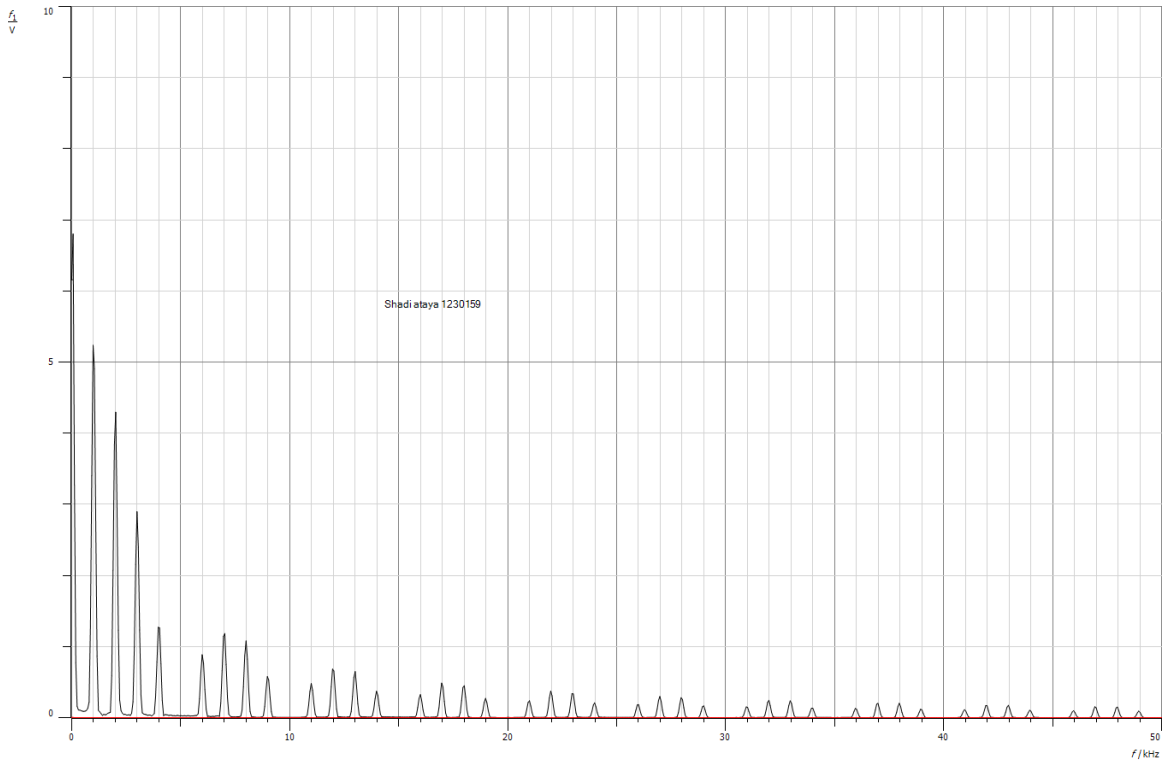


Figure 14: Pulse Train in frequency domain (D = 20%)

We can conclude that the duty cycle affects the locations of the zero crossings in the frequency spectrum: when the duty cycle increases and the pulse becomes wider, the zero crossings move closer to the origin, and when the duty cycle decreases and the pulse becomes narrower, the zero crossings move farther away from the origin. We can notice from the spectrum that the first zero crossing when D = 20% occurs at 5khz also we can calculate it by ($\frac{F}{D} = \frac{1000}{0.2}$ = 5khz) this also means that zero crossing happens each 5khz in the frequency domain, based on the calculations and the figure our result is correct.

Zero crossing when D = 30%/40%/50%/90%

F = 1000 Hz

Based on the previous law we calculate them, and the results are

Table 1: Zero crossing Table

Duty Cycle (%)	20	30	40	50	90
Zero Crossing (kHz)	5	3.3	2.5	2	1.1

- Pulse-train signals require a large transmission bandwidth because their sharp transitions produce many high-frequency harmonics. According to the Nyquist criterion, the bandwidth must be at least twice the highest frequency component in the signal to allow accurate reconstruction without distortion. Therefore, the presence of these high-frequency components forces pulse trains to occupy a wide bandwidth.
- The spectrum of a pulse train consists of an infinite series of harmonics whose amplitudes decrease with increasing frequency, forming a sinc-shaped envelope. This sinc envelope produces multiple zero crossings in the frequency domain. In essence, the pulse train's spectrum is a set of discrete harmonic lines whose magnitudes follow the sinc-shaped envelope determined by the Fourier transform of a single pulse.

Part 2: Characteristics of Pulse Amplitude Modulation (PAM)

In this part, a square signal has been used to sample the message signal. To carry out this experiment, the components have been assembled as shown below.

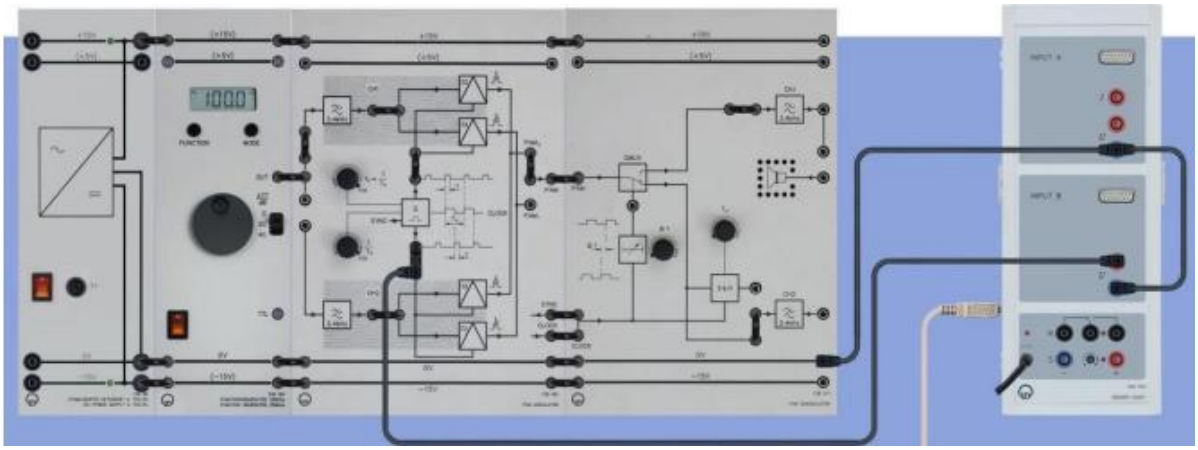


Figure 15: Characteristics of Pulse Amplitude Modulation (PAM) Setup.

In this part the duty-cycle knob (τ/T_p) has been adjusted to its maximum value, and Cassy UA1 has been connected to the clock-generator output (G). The sampling frequency has been set to 5000 Hz by adjusting the clock generator accordingly. An FFT analyser measurement has then been started, and a suitable measuring time has been chosen so that the spectrum becomes clear. The pulse frequency (f_p) has been slowly reduced until the spectral line of the fundamental pulse has appeared at f0=5000, with the second pulse appearing at 3f0=15 kHz and subsequent harmonics following.

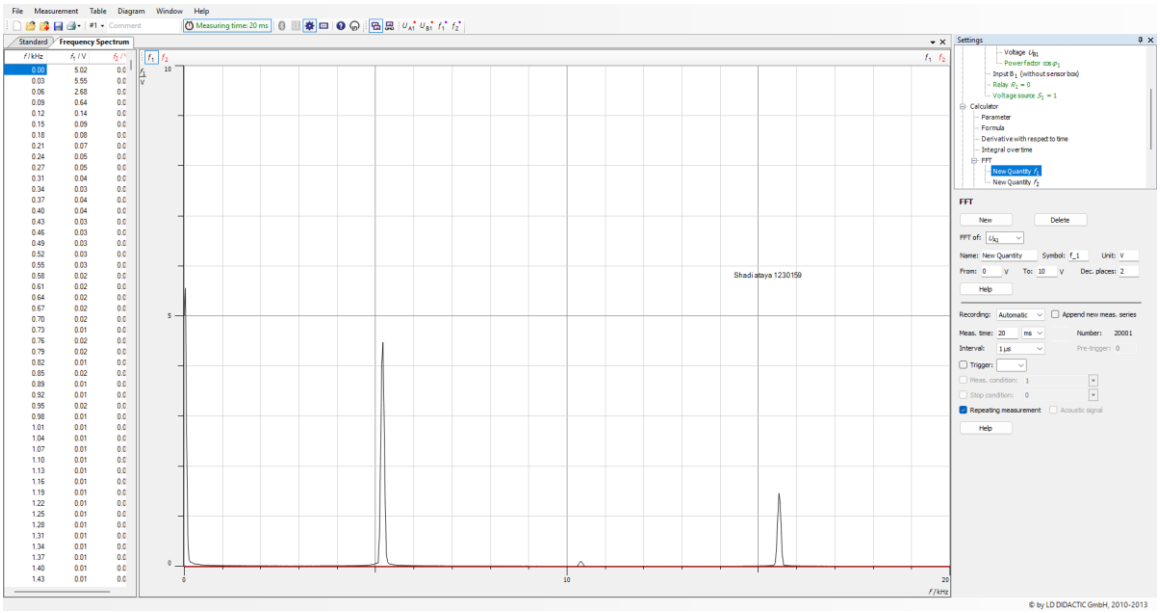


Figure 16: Sampling frequency = 5KHz in frequency domain

Study the effect of the channel filter (CH1):

To study the effect of the filter, a signal of specified amplitude and frequency has been passed through the filter, and the resulting output has been compared with the original signal. The function generator has been set to produce a

sine wave, with a **frequency of 500 Hz** and an **amplitude of Vss = 10 V**

Then Cassy sensor UA1 has been connected to the input message signal, while the Cassy sensor UB1 has been connected to the output of the CH1 filter.

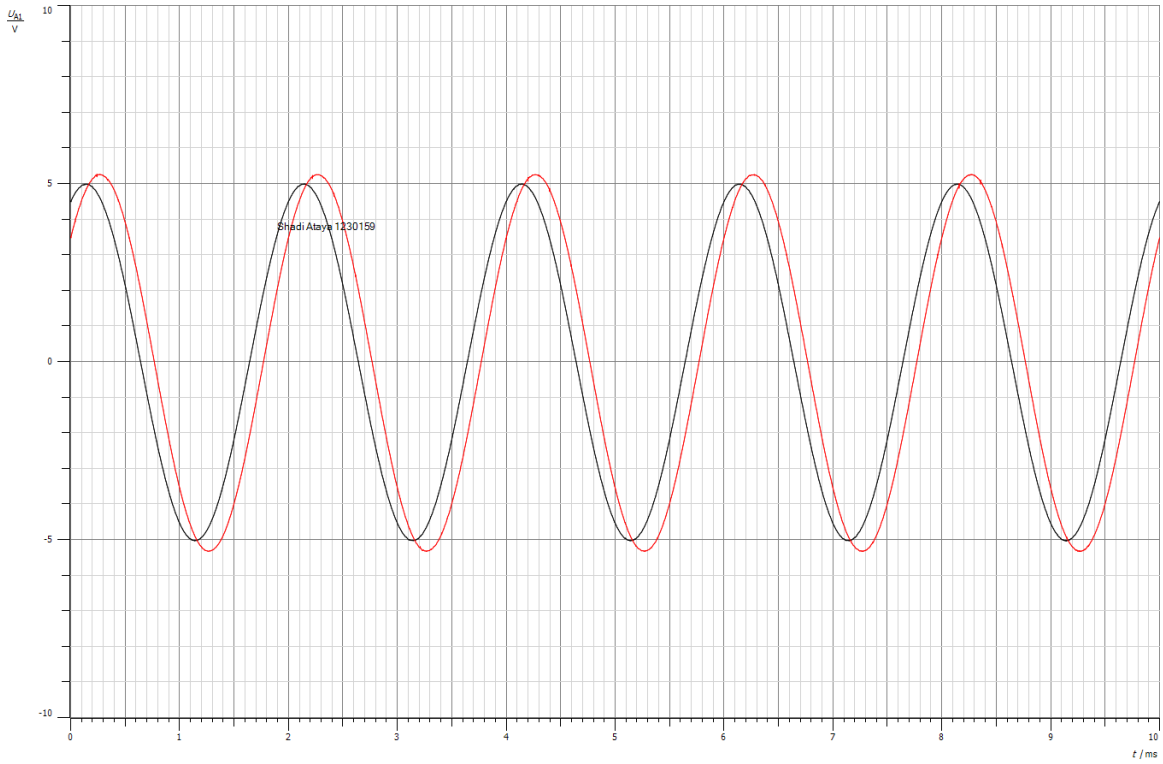


Figure 17: Input and output of the CH1 filter in time domain.

- The filter usually does a phase shift to the original signal while keeping its general shape unchanged. In PAM systems, the filter is important because it removes unwanted high-frequency components and allows only the desired frequencies to pass, producing a cleaner and more accurate signal at the output.
- The gain of the filter at 500Hz

  We can find the gain by dividing the output voltage by the input voltage.

  Gain = Vo/ Vin = $\frac{5.2}{5}$ = 1.04 ,because Gain > 1 the filter also amplifies the signal

Display the Pulse Amplitude Modulated signal s(t) of PAM1 in the time domain.

The function generator has been set to produce a sine wave at 500 Hz with a peak-to-peak voltage of 10 V. The output signal of the CH1 filter has been connected to the Cassy sensor UA1. The PAM1 output has been connected to the Cassy sensor UB1.

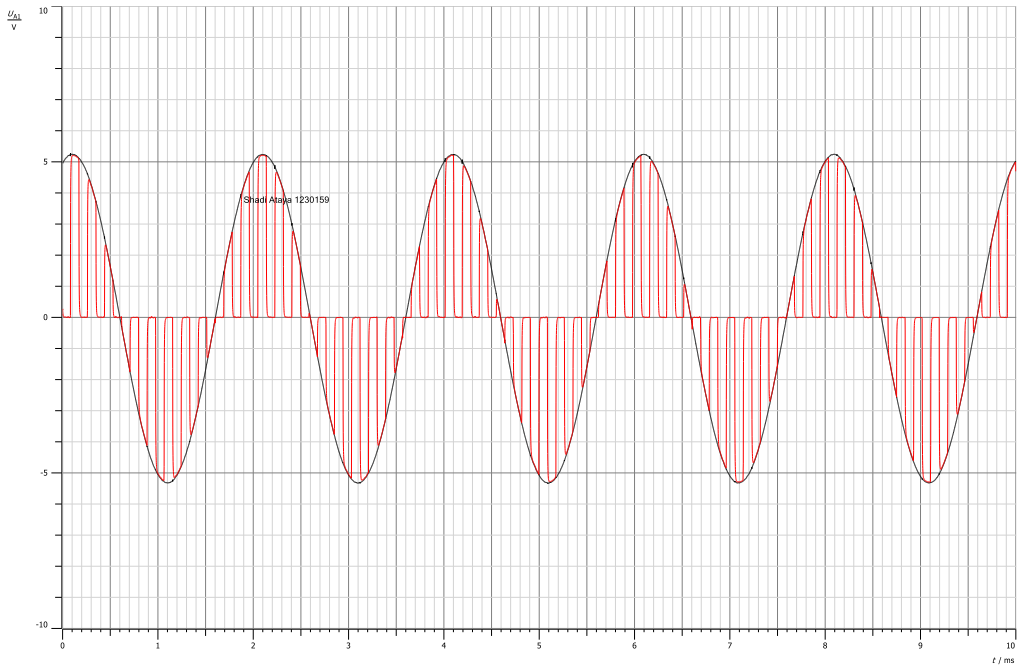


Figure 18: PAM1 output with frequency = 500 in time domain

Determining the effect of the duty cycle on the PAM in the time domain:

The duty cycle of the clock generator has been reduced to the minimum value.

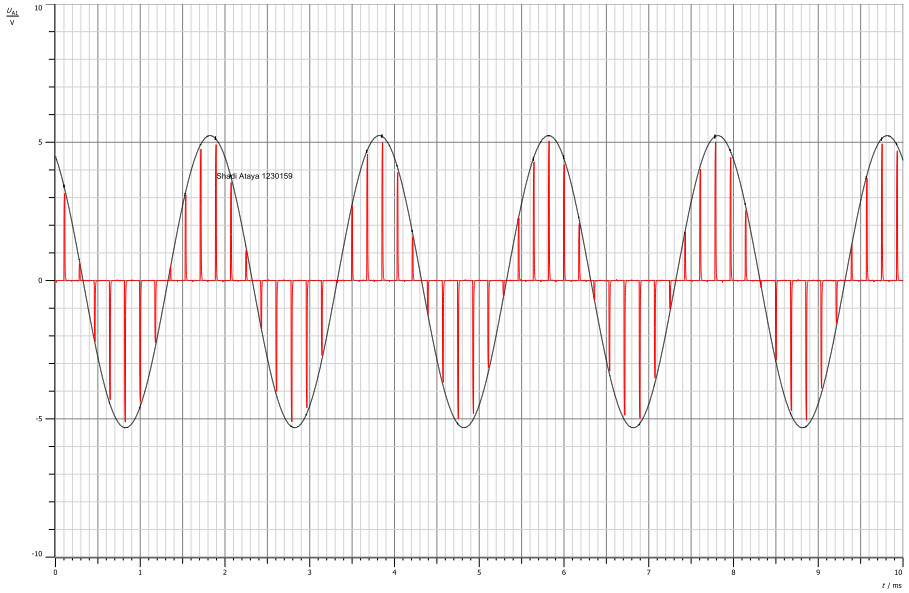


Figure 19: PAM1 output with frequency = 500 in and minimum duty cycle time domain

- In the time domain, reducing the duty cycle of the clock generator to its minimum in natural sampling PAM produces very narrow pulses, increasing the gaps between them, which makes the sampled signal sparser, less accurately follows the input waveform, and increases distortion.

Display the Pulse Amplitude Modulated signal s(t) of PAM1 in the Frequency domain:

The same connections used to study the time domain of PAM1 have been used. The duty cycle of the clock generator has been set to the maximum value. The Cassy sensor UA1 has been connected to the clock-generator output.

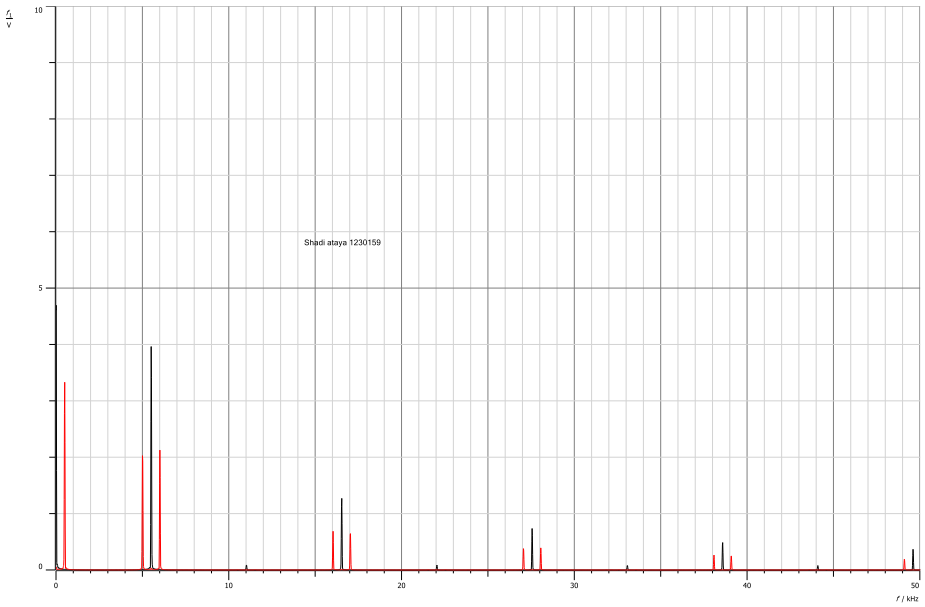


Figure 20: PAM1 output with frequency = 500 in frequency domain

Determining the effect of the message frequency on the PAM in the Freq domain:

The frequency has been changed to 1khz then 2khz.

First case (1KHz)

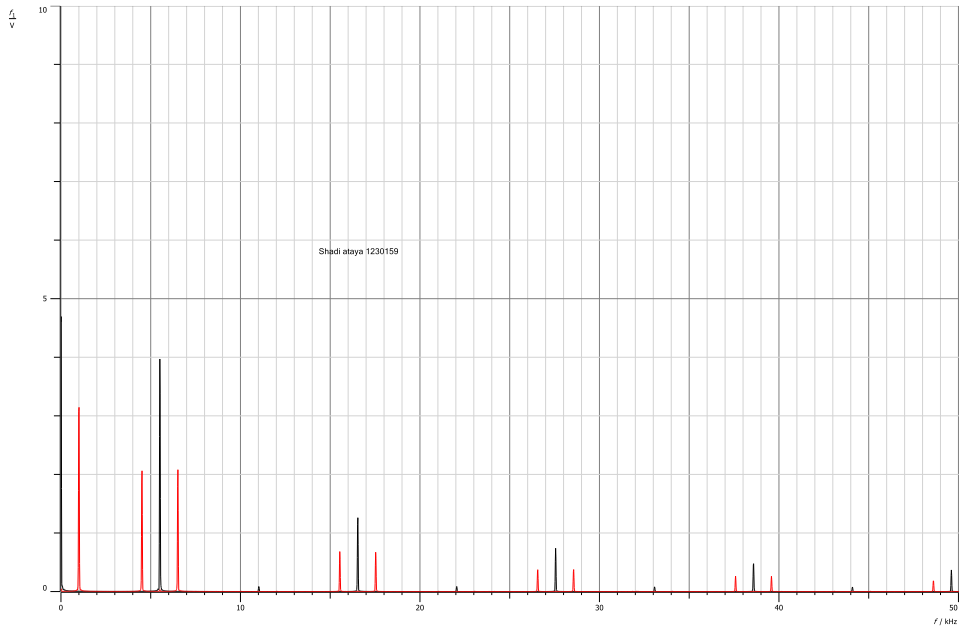


Figure 21: PAM1 output with frequency = 1000 in frequency domain

Second case (2KHz)

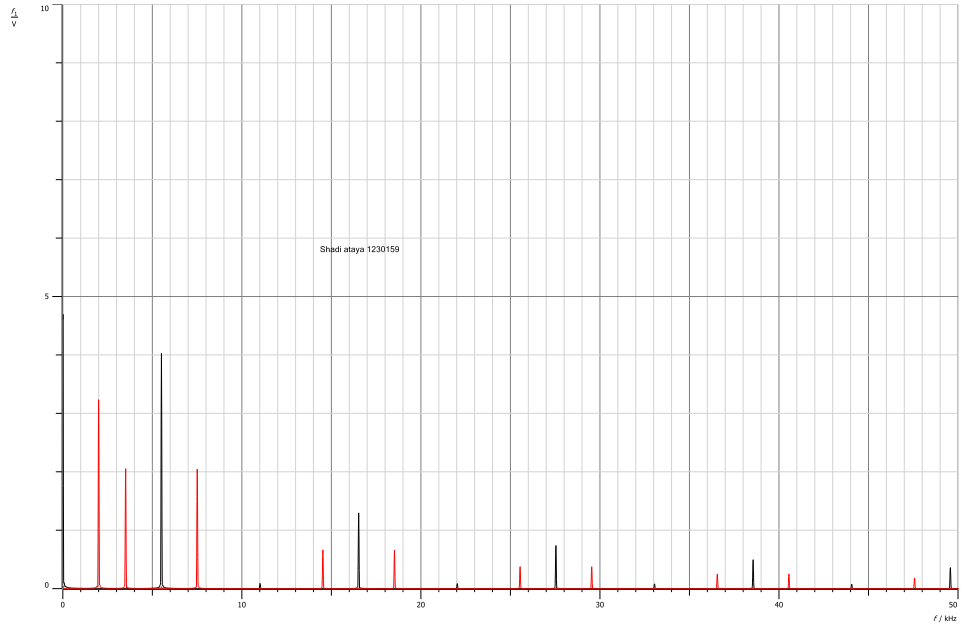


Figure 22: PAM1 output with frequency = 2000 in frequency domain

- In natural sampling PAM, increasing the message frequency from 1 kHz to 2 kHz shifts the spectral components to higher frequencies within each replica, causing greater attenuation due to the pulse's sinc envelope and making the higher-frequency components weaker in the spectrum.

Determining the effect of the duty cycle on the PAM1 in the Freq domain:

The message frequency has been set back to 500 Hz. The duty cycle of the clock generator has been changed to 10% and 30%, after finished It has been returned to 50%.

First case (D = 10%)

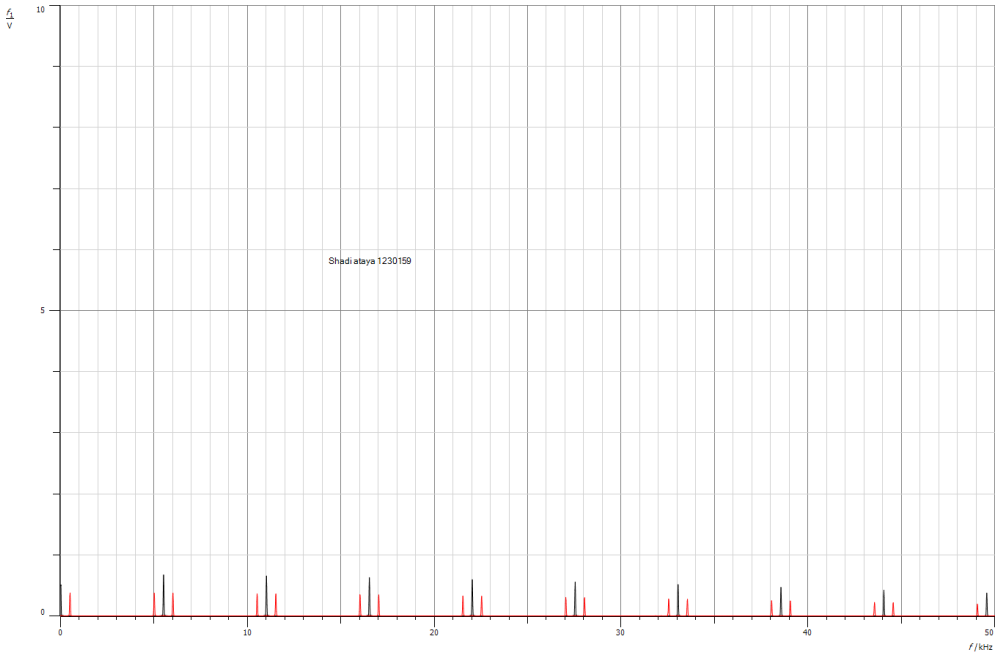


Figure 23: PAM1 output with Duty cycle 10% in frequency domain.

Second case (D = 30%)

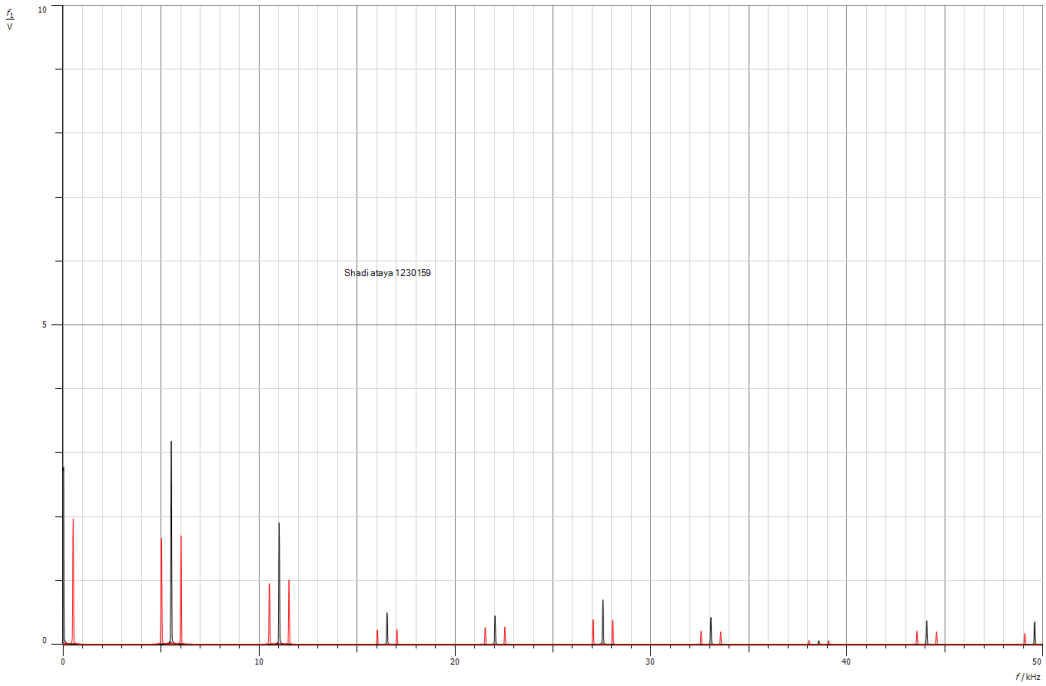


Figure 24: PAM1 output with Duty cycle 30% in frequency domain.

In natural sampling PAM, changing the duty cycle of the clock generator to 10% and 30% affects the spectral width, as the narrower pulses produced by a 10% duty cycle contain more high-frequency components and therefore increase the overall bandwidth, while the wider pulses from a 30% duty cycle concentrate more energy at lower frequencies, reducing the bandwidth and tightening the spectral envelope around each replica.

Display the Pulse Amplitude Modulated signal s(t) of PAM2 in the time domain:

The function generator has been set to produce a sine wave at 500 Hz with a peak-to-peak voltage of 10 V. The Cassy sensor UA1 has been connected to the output signal of the CH1 filter, and the Cassy sensor UB1 has been connected to the PAM2 output.

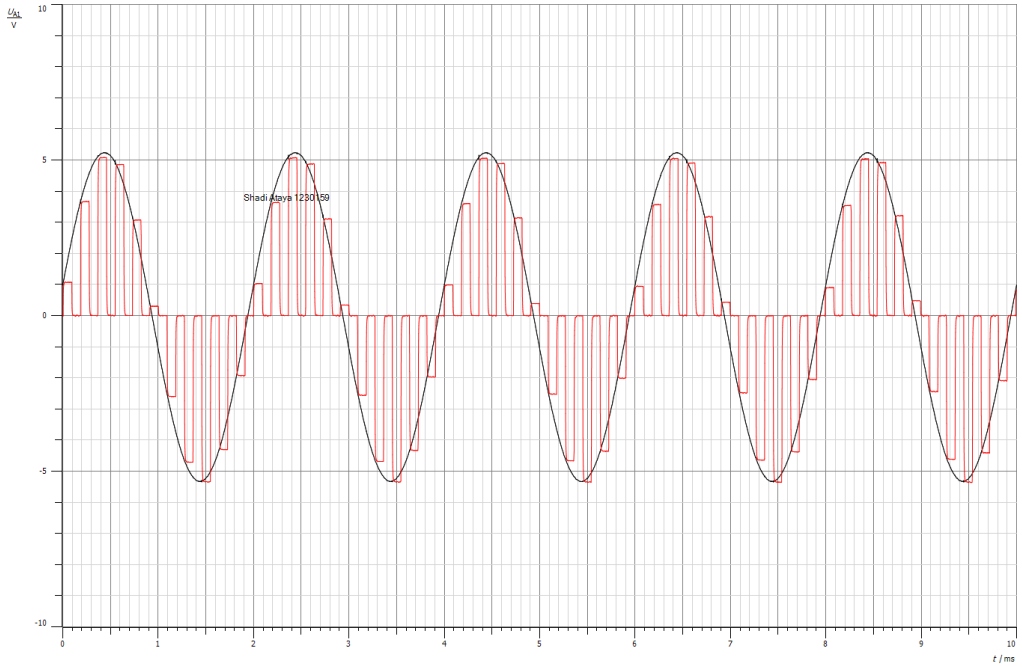


Figure 25: PAM2 output with frequency = 500 in time domain

Determining the effect of the duty cycle on the PAM2 in the time domain:

The duty cycle of the clock generator has been reduced to the minimum value.

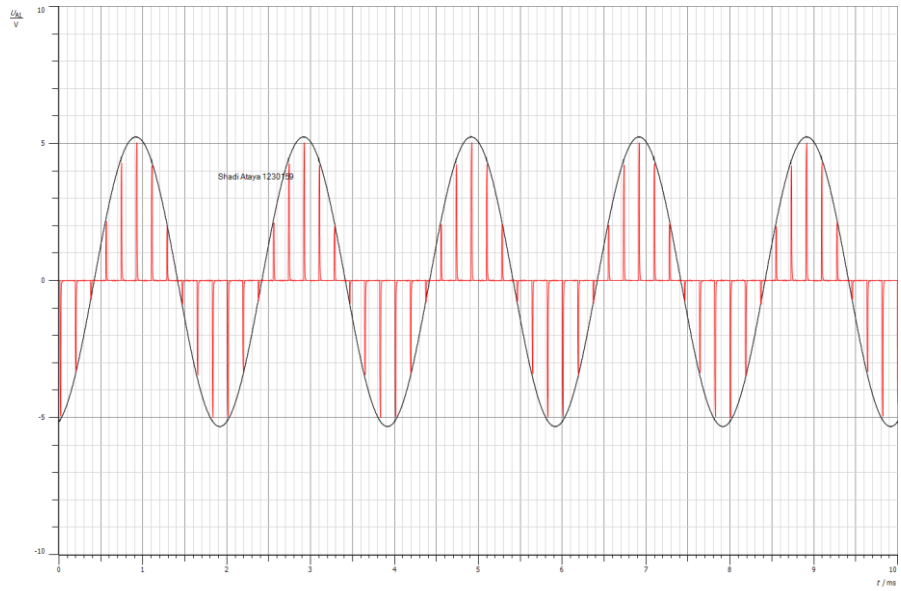


Figure 26: PAM2 output with frequency = 500 in and minimum duty cycle time domain

- In flat-top PAM, reducing the duty cycle to its minimum shortens the flat portion of each pulse, causing each sample to be held for a very brief time; this makes the waveform more discrete, increases distortion, and reduces the accuracy with which the held levels represent the original signal in the time domain.

Display the Pulse Amplitude Modulated signal s(t) of PAM2 in the Frequency domain:

The same connection that was used to study the time domain of PAM2 has been used. The duty cycle of the clock generator has been set to the maximum value, and the Cassy sensor UA1 has been connected to the clock-generator output.

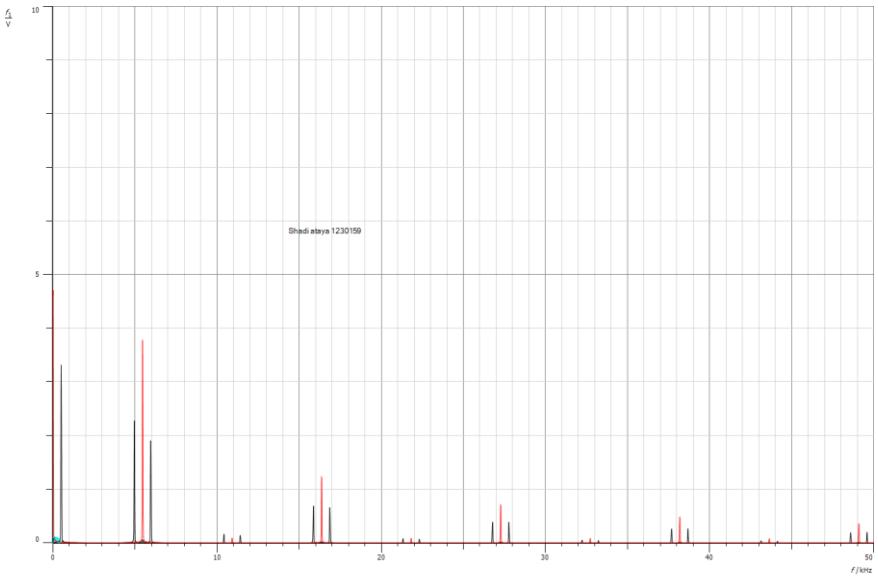


Figure 27: PAM2 output with frequency = 500 and maximum duty cycle in frequency domain

Determining the effect of the message frequency on the PAM in the Freq domain:

The message frequency has been changed to 1 kHz and 2 kHz.

First case (1KHz)

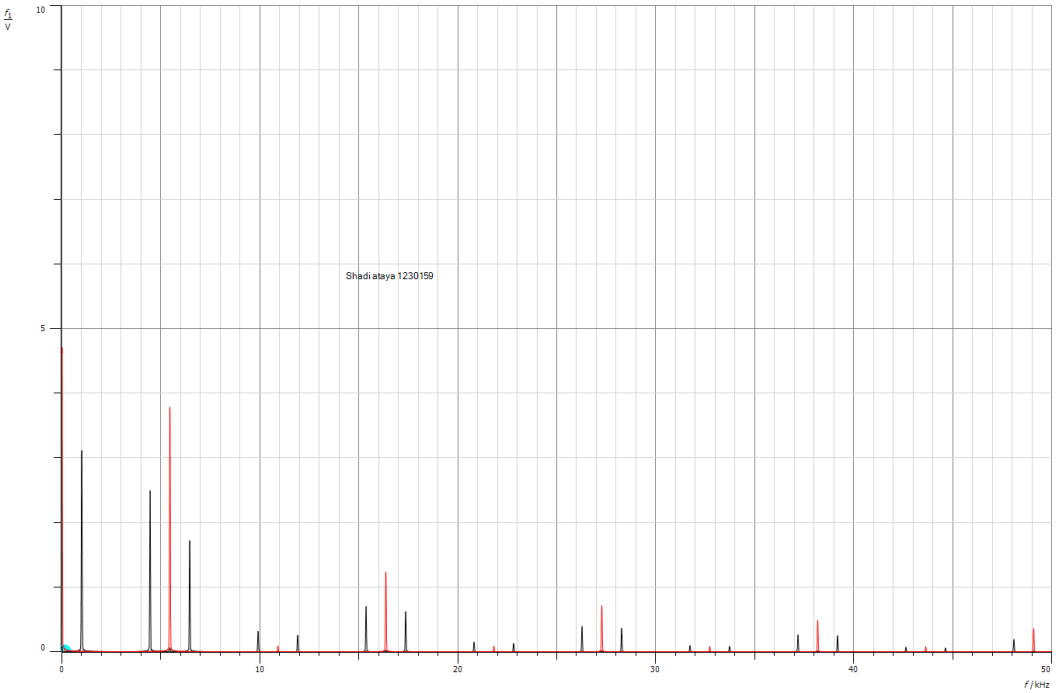


Figure 28: PAM2 output with frequency = 1000 in frequency domain

Second case (2KHz)

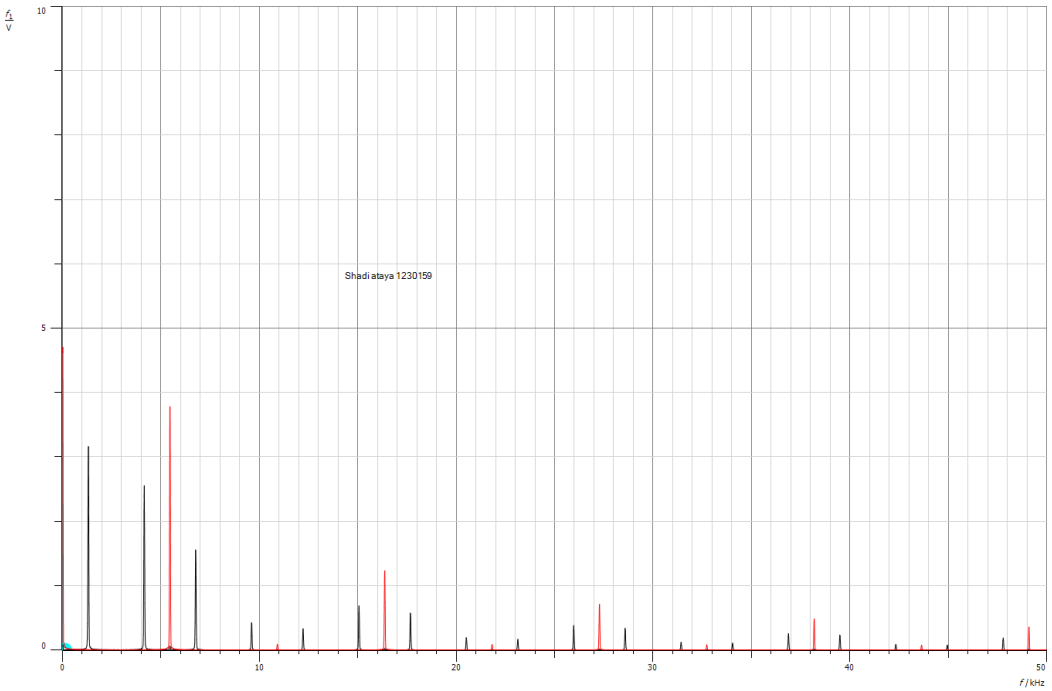


Figure 29: PAM2 output with frequency = 2000 in frequency domain

- In flat-top PAM, changing the message frequency to 1 kHz and 2 kHz shifts the message components to higher frequencies within each spectral replica, and the higher 2 kHz component experiences greater attenuation because flat-top sampling introduces a hold effect that multiplies the spectrum by a sinc function, causing higher-frequency components to weaken more in the frequency domain.

Determining the effect of the duty cycle on the PAM2 in the Freq domain:

The message frequency has been set back to 500 Hz. The duty cycle of the clock generator has been changed to 10% and 30%, after finished It has been returned to 50%.

First case (D = 10%)

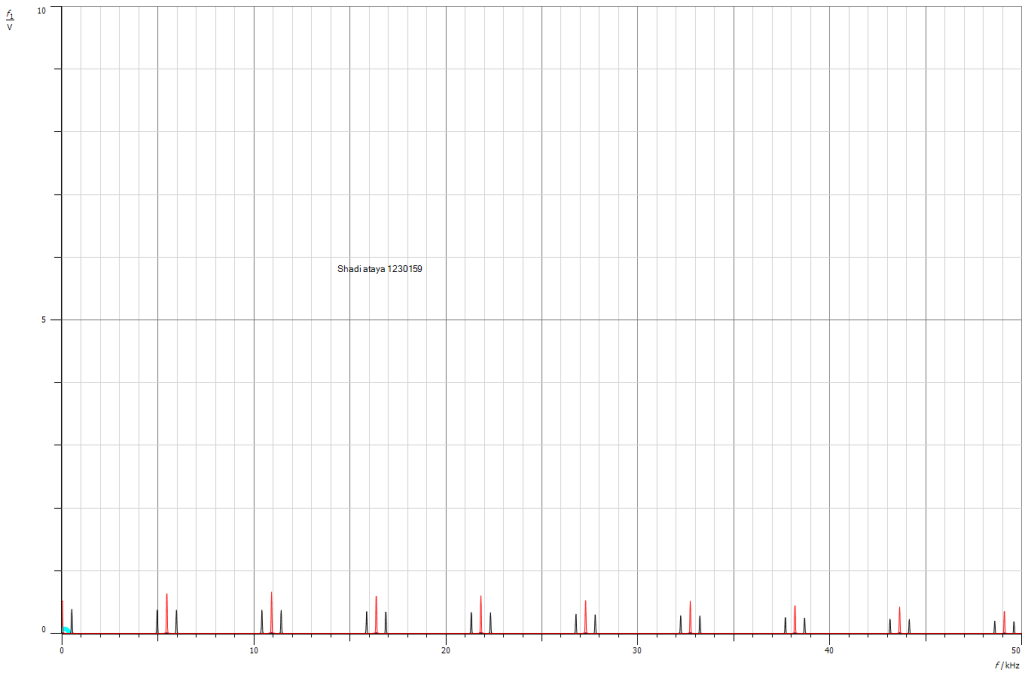


Figure 30: PAM2 output with Duty cycle 10% in frequency domain.

Second case (D = 30%)

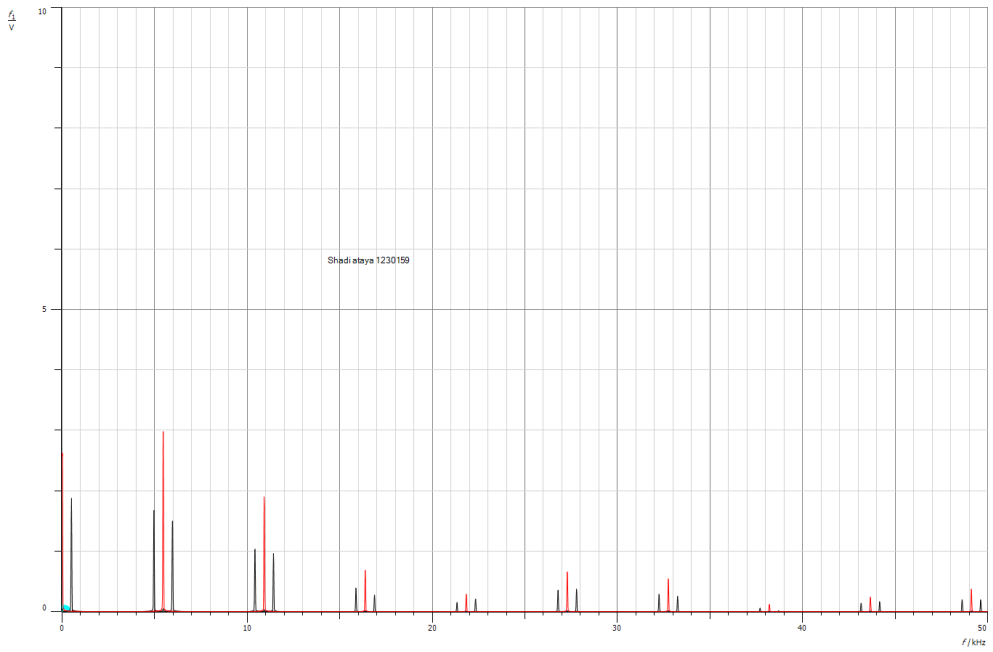


Figure 31: PAM2 output with Duty cycle 10% in frequency domain.

In flat-top PAM, changing the duty cycle to 10% and 30% alters the spectrum by modifying the pulse width, where the narrower 10% duty-cycle pulses broaden the spectral envelope and

increase the bandwidth, while the wider 30% pulses narrow the envelope, reduce the bandwidth, and lessen high-frequency attenuation caused by the hold (sinc) effect.

Part 3: Characteristics of Pulse Amplitude Demodulation

Display the message signal and the demodulated signal using PAM1:

The clock generator frequency is set to its maximum value, while the function generator is adjusted to produce a 500 Hz sine wave with a 10 V peak-to-peak amplitude. The Cassy sensor UA1 is connected to the input signal of the CH1 filter, and the Cassy sensor UB1 is connected to the output of the CH1 demodulator filter.

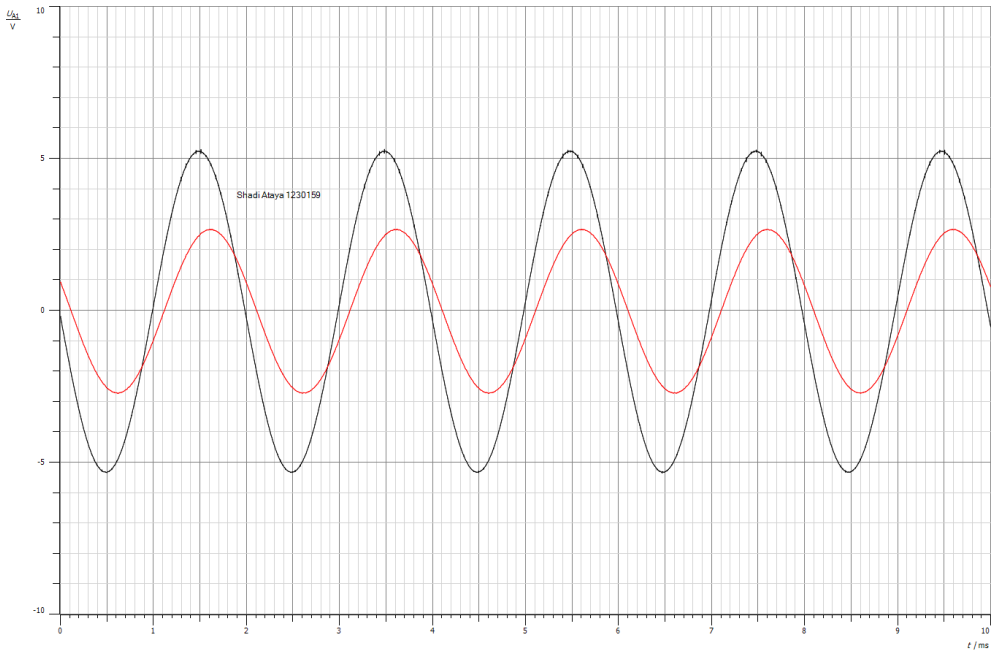


Figure 32: Demodulation output in time domain.

Determining the effect of the duty cycle on the PAM in the time domain:

The duty cycle of the clock generator is changed to 10% and 30%, respectively.

First Case (D = 10%)

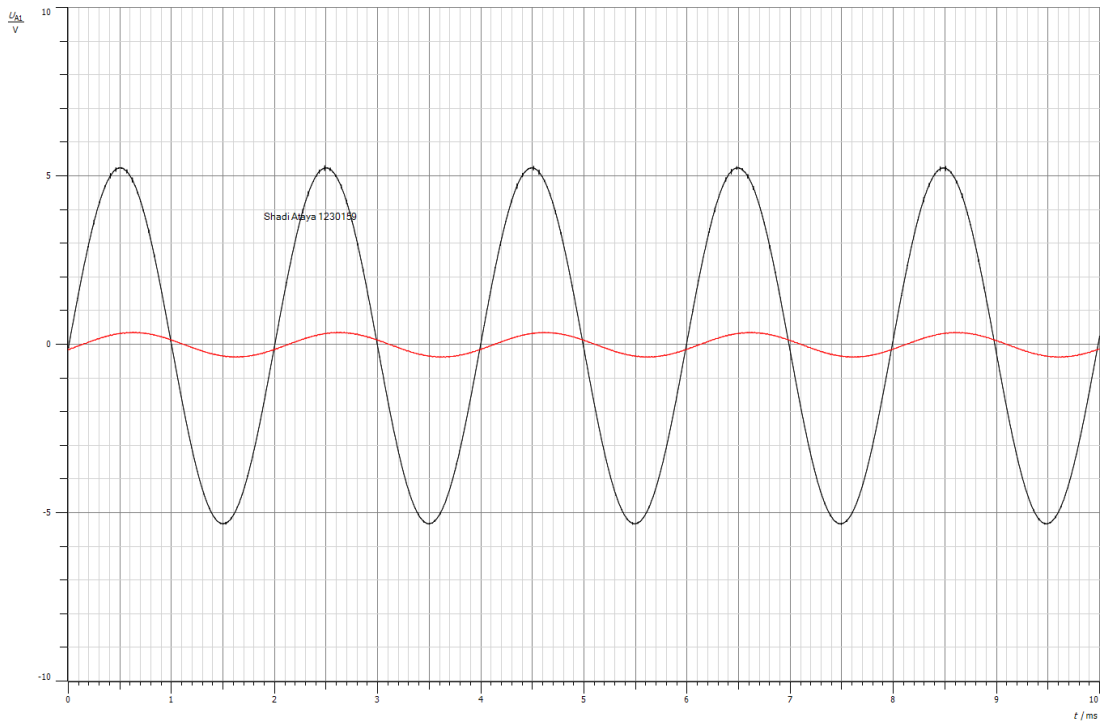


Figure 33: Demodulation output with duty cycle 10% in time domain.

Second Case (D = 30%)

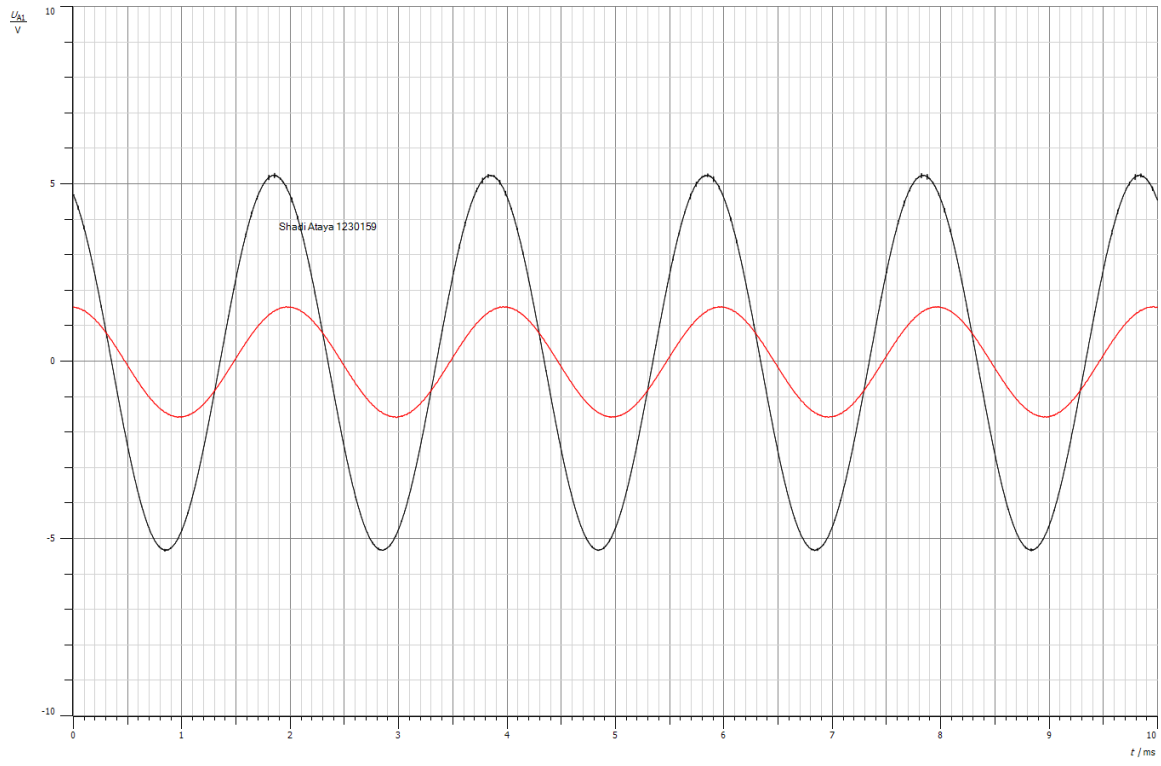


Figure 34: Demodulation output with duty cycle 30% in time domain.

- As shown, in Pulse Amplitude Modulation (PAM), altering the duty cycle affects demodulation efficiency and signal fidelity in the time domain. Increasing the duty cycle reduces distortion by making the demodulator less sensitive to rapid signal variations, but it requires higher sampling rates and wider bandwidths. For example, a 30% duty cycle improves demodulation efficiency compared to 10%, enabling more accurate signal reconstruction. Conversely, decreasing the duty cycle shortens the ON periods of the pulses, which can increase signal errors in time-division multiplexing.

Part 4: Aliasing in the Time and the Frequency Domains

Display the modulated signal and the demodulated signal using PAM1 in the time domain:

The clock generator frequency is set to 5000 Hz, and the duty cycle is adjusted to 50%. The function generator is set to a sine wave with a frequency of 3000 Hz and VSS = 5 V. For the modulated signal CASSY sensor UA1 is connected to the output signal of the CH1 filter, and the CASSY sensor UB1 is connected to the PAM1 modulator.

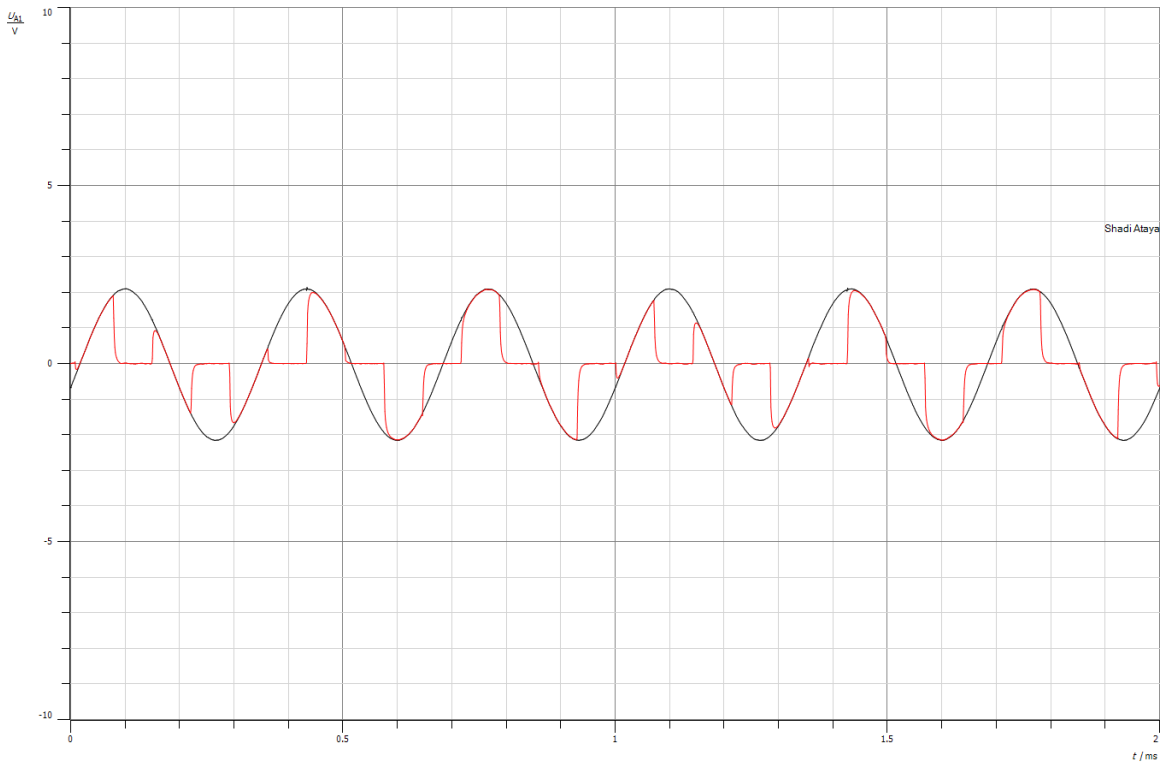


Figure 35: Modulated Signal using PAM1 with aliasing in time domain.

Now for the demodulated signal the CASSY sensor UB1 is connected to the output of the CH1 demodulator filter.

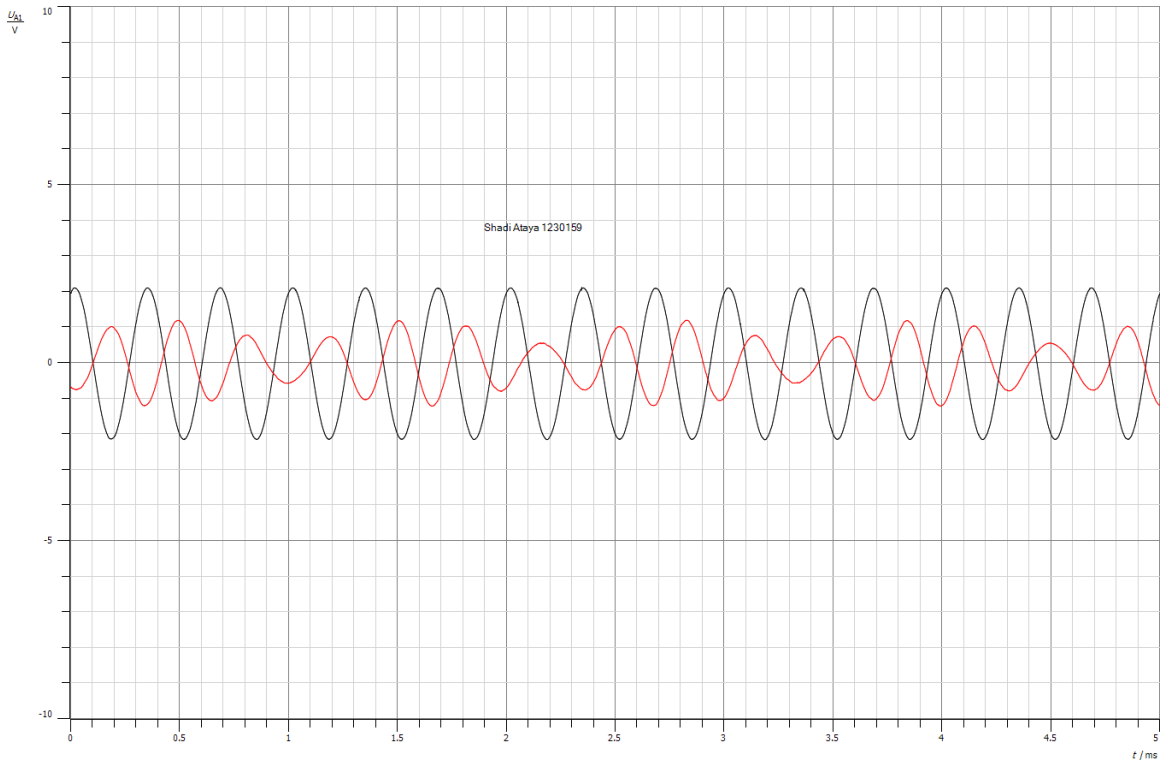


Figure 36: Demodulated Signal using PAM1 with aliasing in time domain.

- When the sampling frequency is 5000 Hz and the message frequency is 3000 Hz, the Nyquist rate is violated because the sampling frequency is not at least twice the message frequency. As a result, aliasing occurs in the time domain. The PAM modulation signal becomes distorted, since the samples cannot follow the fast variations of the 3000 Hz waveform, causing the sampled envelope to appear as a slower, incorrect signal. After demodulation, the original message cannot be recovered; instead, the filter outputs the aliased component, producing a lower-frequency waveform that does not match the true message.

Display the modulated and the demodulated signals using PAM1 in the frequency domain:

The same message settings used for studying the time domain of PAM1 are applied. The CASSY sensor UA1 is connected to the output of the clock generator, and the CASSY sensor UB1 is connected to the PAM1 modulator.

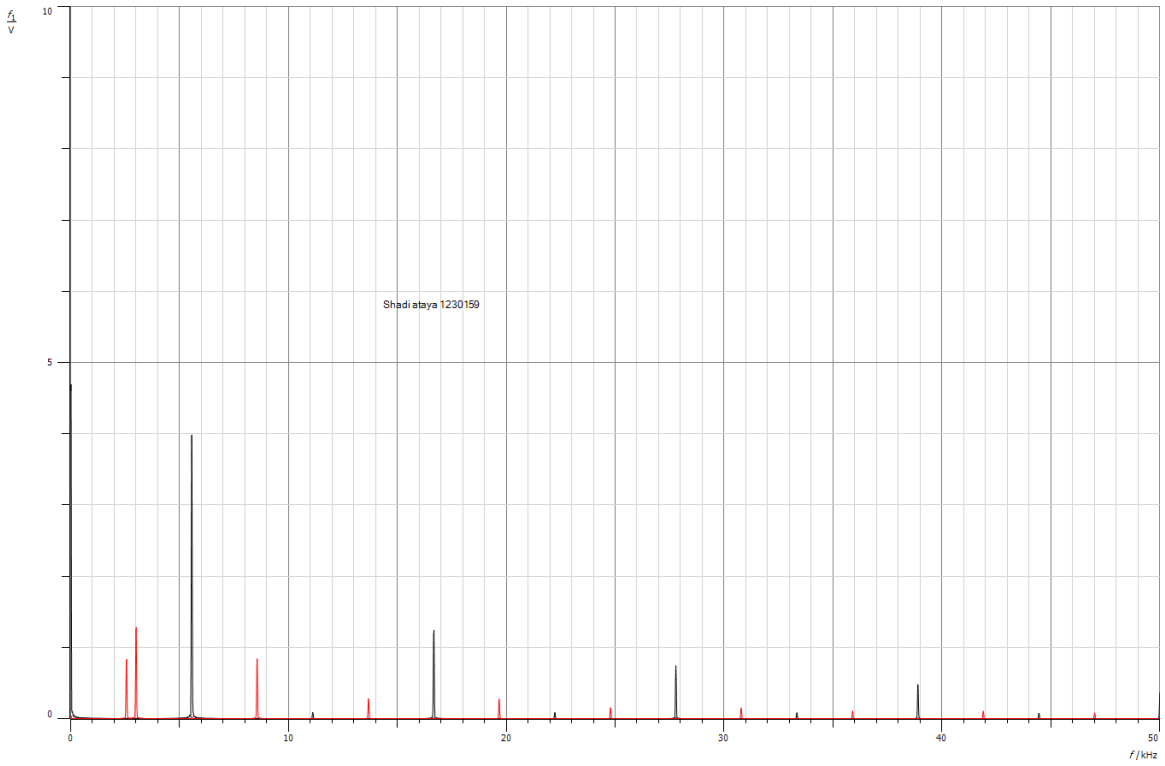


Figure 37: Modulated signal with aliasing in frequency domain.

Now Cassy sensor UB1 has been connected to the output of the demodulator filter of CH1.

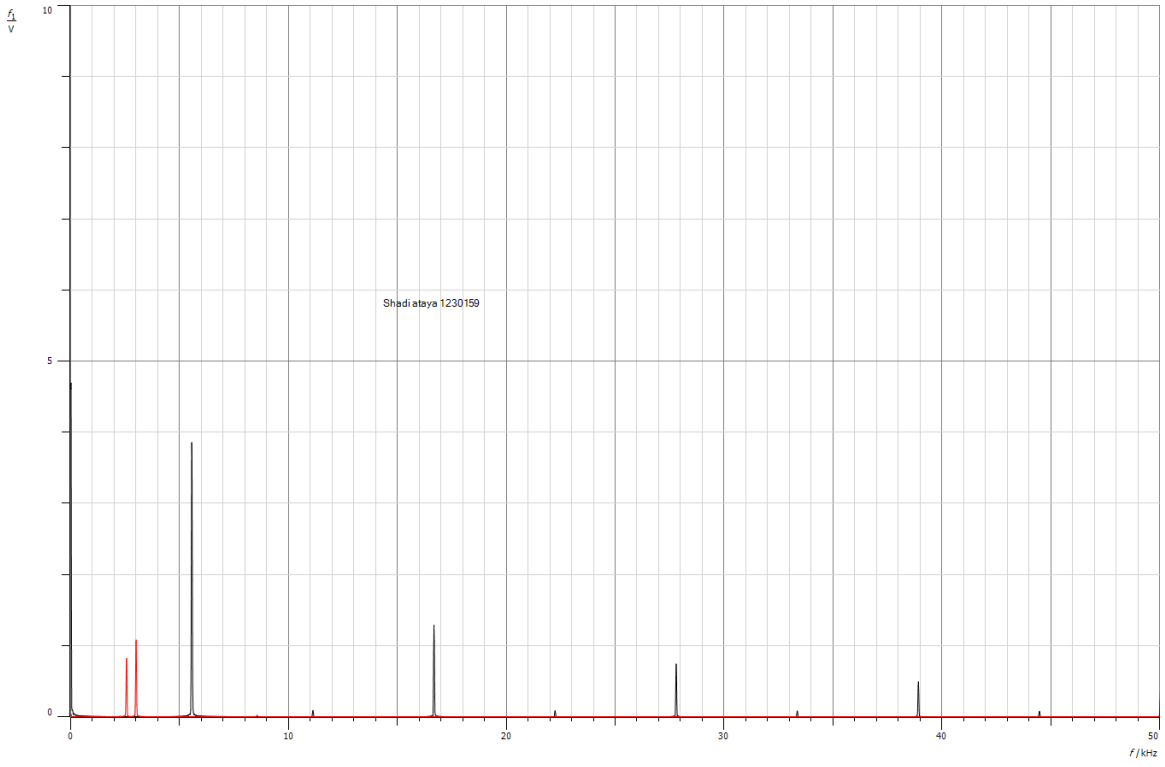


Figure 38: Demodulated signal with aliasing in frequency domain.

Part 5: PAM Time Multiplex

To carry out this experiment, the components are assembled as shown in the figure below.

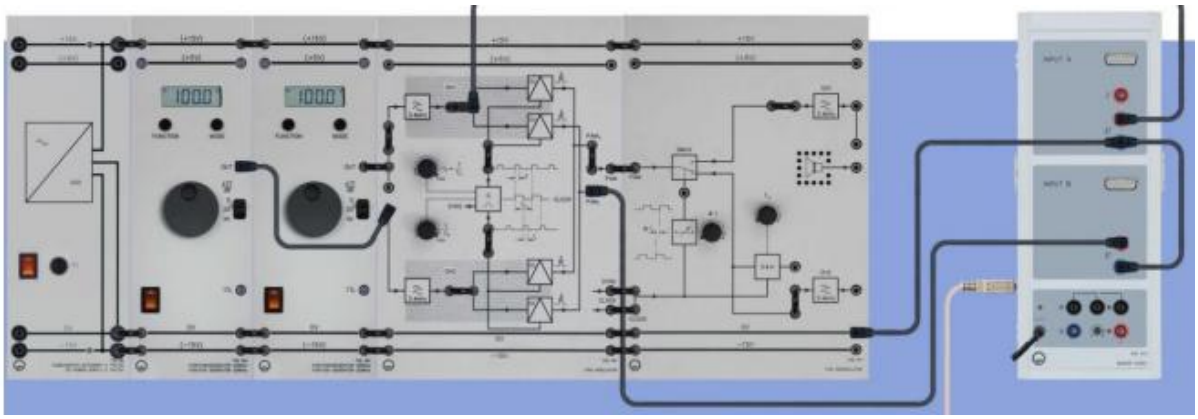


Figure 39: PAM Time Multiplex setup.

Display the time characteristic of the time multiplex signal.

The sampling frequency fP is set to its maximum, and the duty cycle is adjusted to its maximum value. The first function generator is configured to produce a triangle waveform with fM1 = 200 Hz and a voltage swing VSS = 5 V. The second function generator is set to a sine waveform with fM2 = 300 Hz and a voltage swing VSS = 10 V. The CASSY sensor UA1 is connected to the triangle message on channel CH1, while the CASSY sensor UB1 is connected to the output of the PAM1 modulator. For demodulation observations, UA1 and UB1 are then connected to the outputs of the CH1 and CH2 demodulator filters, respectively.

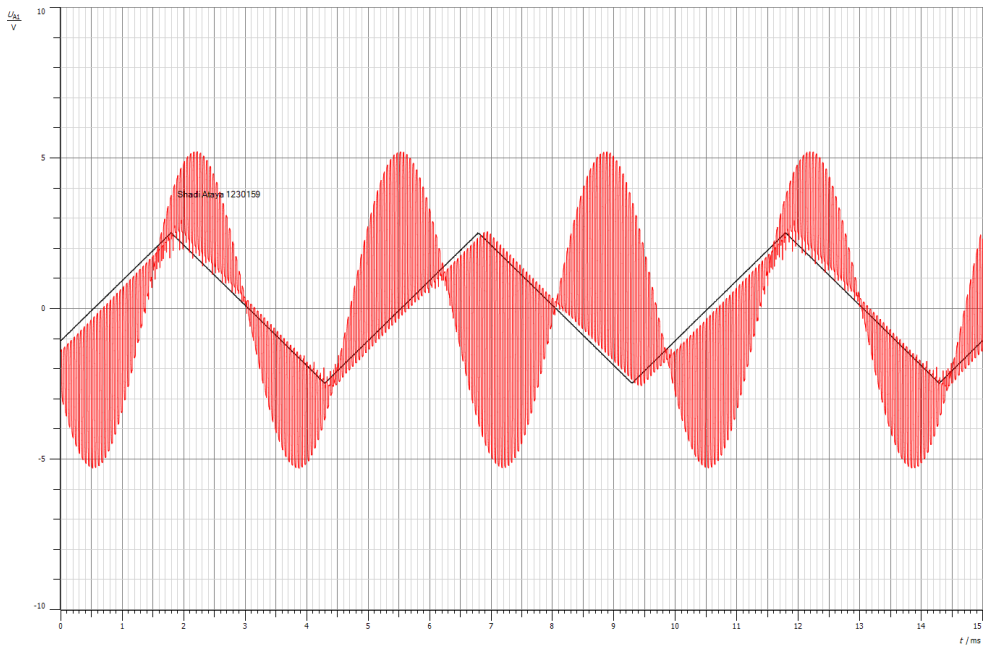


Figure 40: Triangular message signal with the PAM1 output in time domain.

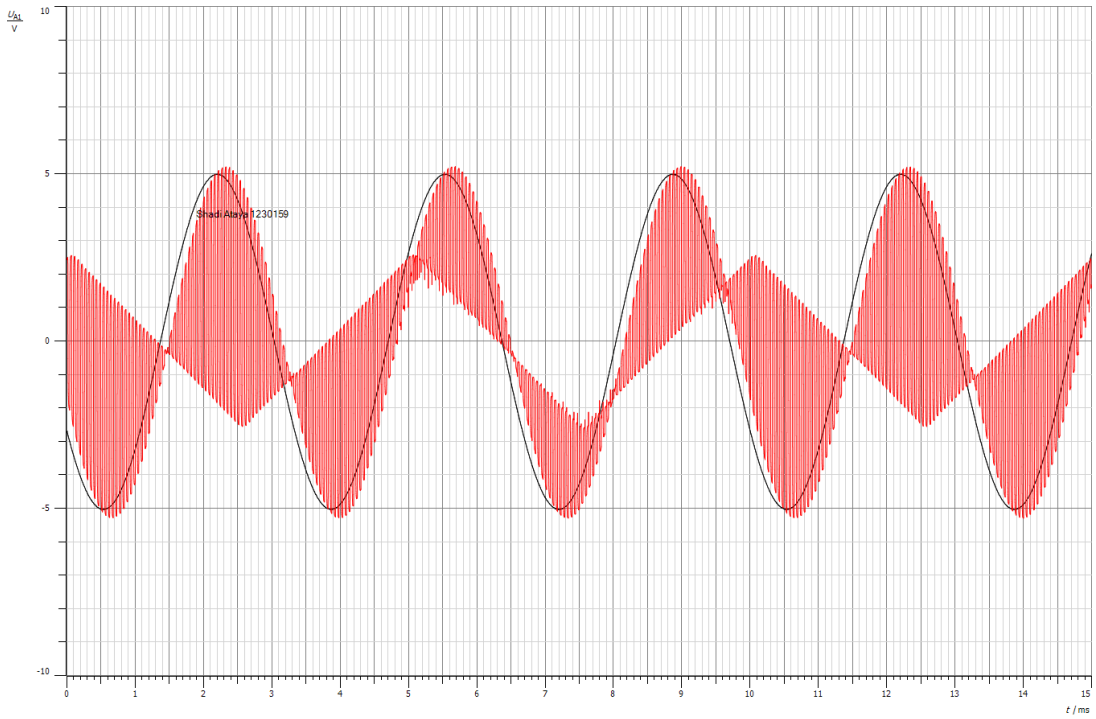


Figure 41: Sine message signal with the PAM1 output in time domain.

Cross Talk (Δt) left/middle.

When synchronization is not maintained within the system, it leads to cross talk. This results in a situation where a portion of the first message $m1(t)$ appears at the output intended for the second message signal $m2(t)$, and vice versa.

By keeping the same connections of the message signals, function generators as: Triangle, fM1 = 200 Hz, VSS = 5 V, Sine, fM2 = 300 Hz, VSS = 10 V.

The figures below show the modulated and demodulated signals in the time domain, showing the changes caused by the phase shift Δt.

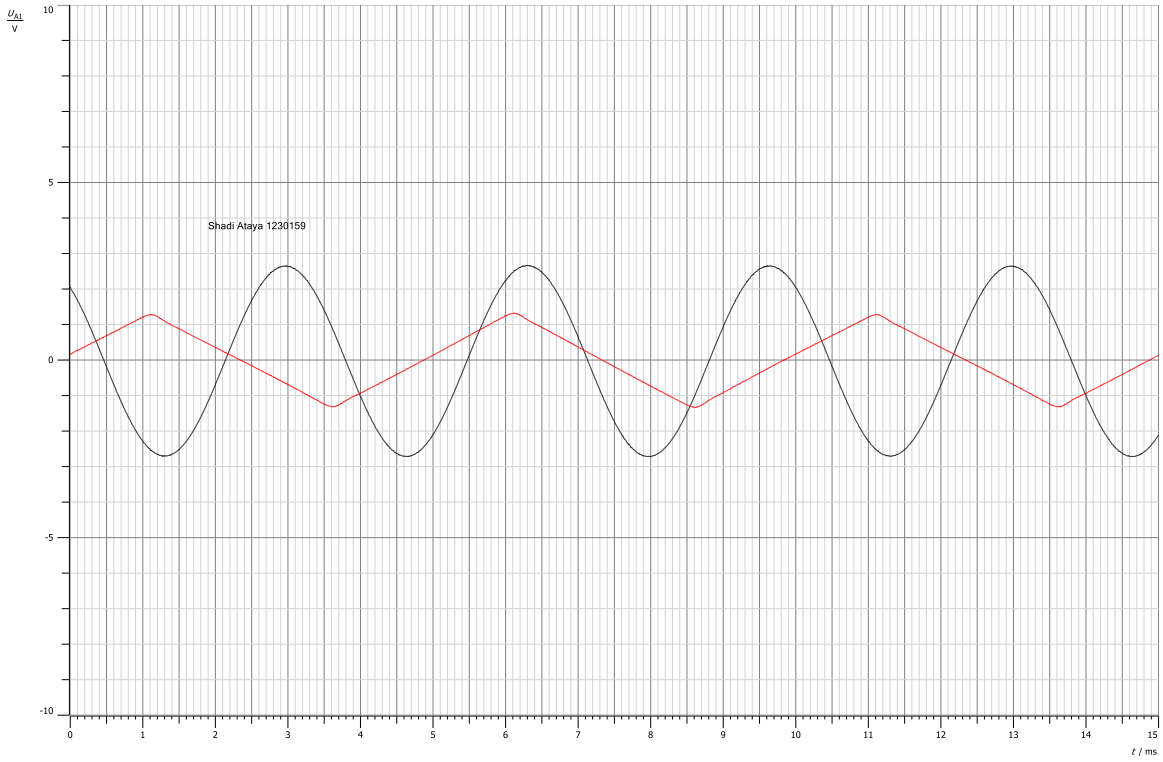


Figure 42: Message Signals before changes phase shift Δt (No crosstalk effect) in time domain.

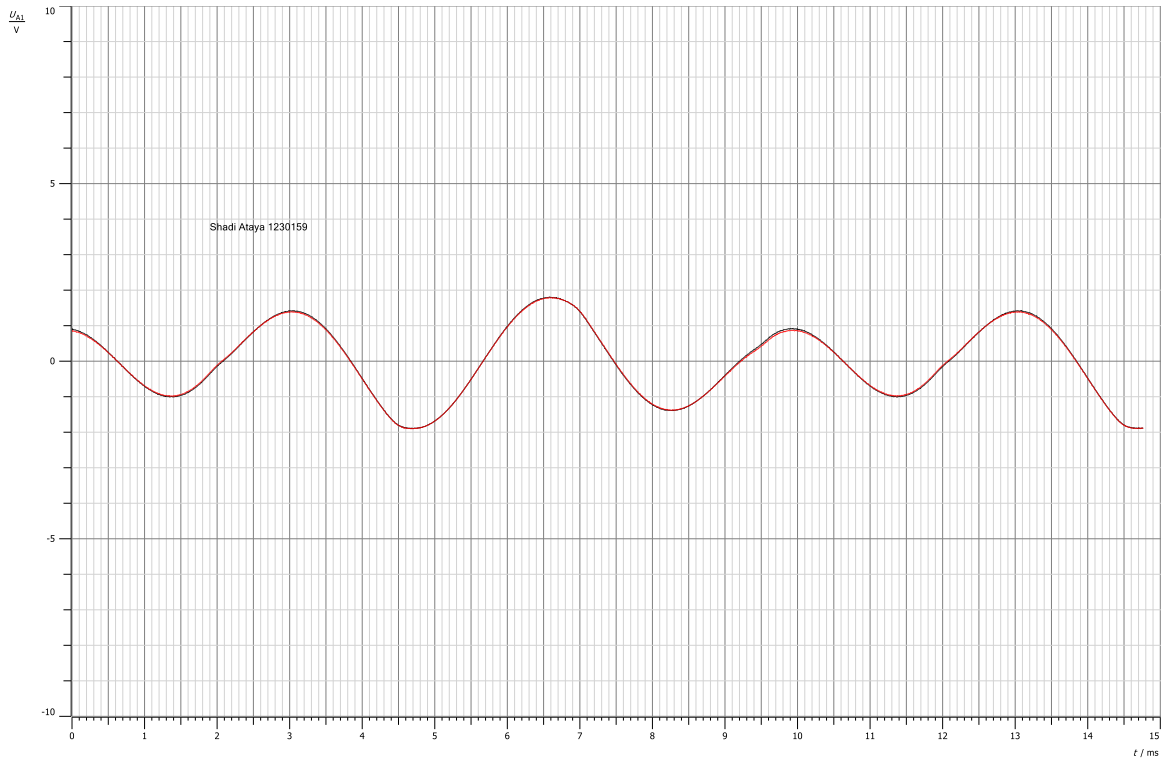


Figure 43: Message Signals after changes phase shift Δt (crosstalk effect) in time domain.



Figure 44: Message Signals after changes phase shift Δt (crosstalk) in frequency domain.

- Crosstalk causes distortion, amplitude errors, interference, and loss of signal quality — the effect changes depending on the phase shift between the signals.

After that, the first function generator has been set to a Triangle waveform with a frequency of **fM1 = 200 Hz** and a **VSS = 10 V**. The second function generator has been set to a Sine waveform with a frequency of **fM2 = 300 Hz** and a **VSS = 10 V**. The effect of cross talk is shown in the following figures, where the phase shift **Δt** is varied and its influence is observed in both the time and frequency domains.

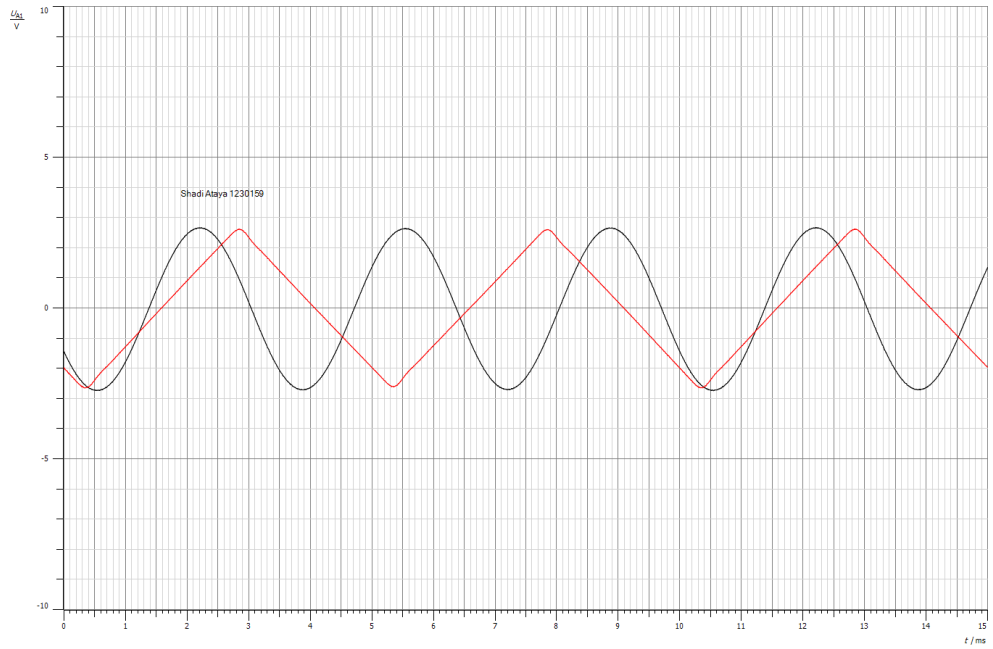


Figure 45: Message Signals before changes phase shift Δt (No crosstalk effect) in time domain.

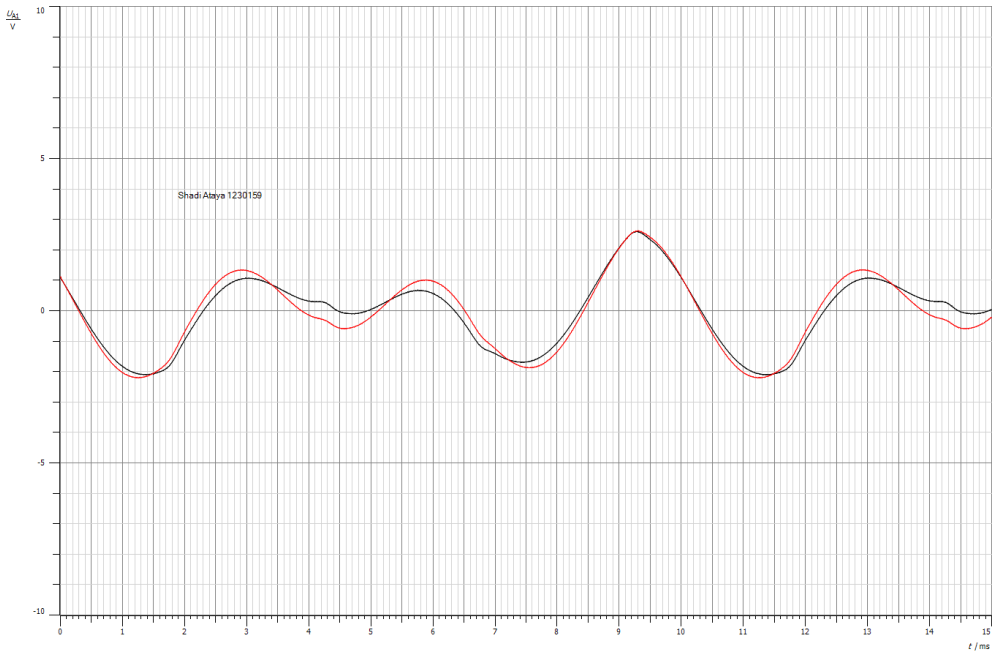


Figure 46: Message Signals after changes phase shift Δt (crosstalk effect) in time domain.



Figure 47: Message Signals after changes phase shift Δt (crosstalk effect) in frequency domain.

- The cross effect here also causes distortion, amplitude errors, interference, and loss of signal quality — the effect changes depending on the phase shift between the signals.

Conclusion

In conclusion, this experiment provided a comprehensive understanding of Pulse Amplitude Modulation (PAM) and its role in communication systems. The theoretical concepts of pulse train spectra, sampling, Nyquist rate, and aliasing were verified through practical measurements, allowing the relationship between sampling parameters and signal quality to be clearly observed. The effects of duty cycle, sampling frequency, and filtering on the PAM waveform were analysed in both the time and frequency domains, demonstrating how these factors influence signal shape, bandwidth, and distortion.

The experiment also explored the operation of Time Division Multiplexing (TDM), where two message signals—a triangular wave and a sine wave—were successfully multiplexed. This allowed for direct observation of the multiplexed signal, and the conditions under which cross talk occurs when synchronization is not maintained. Through these observations, the importance of timing accuracy and proper sampling rate selection became evident, especially in avoiding aliasing and preventing interference between channels.

Overall, the experiment effectively connected theoretical knowledge with practical implementation. It reinforced key communication concepts, demonstrated the behaviour of real systems under different sampling conditions, and highlighted the critical role of synchronization and filtering in achieving accurate modulation and demodulation.

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